

Experiment 3: Creation of a Simple Network Topology Using Open-Source Network Virtualization Tools

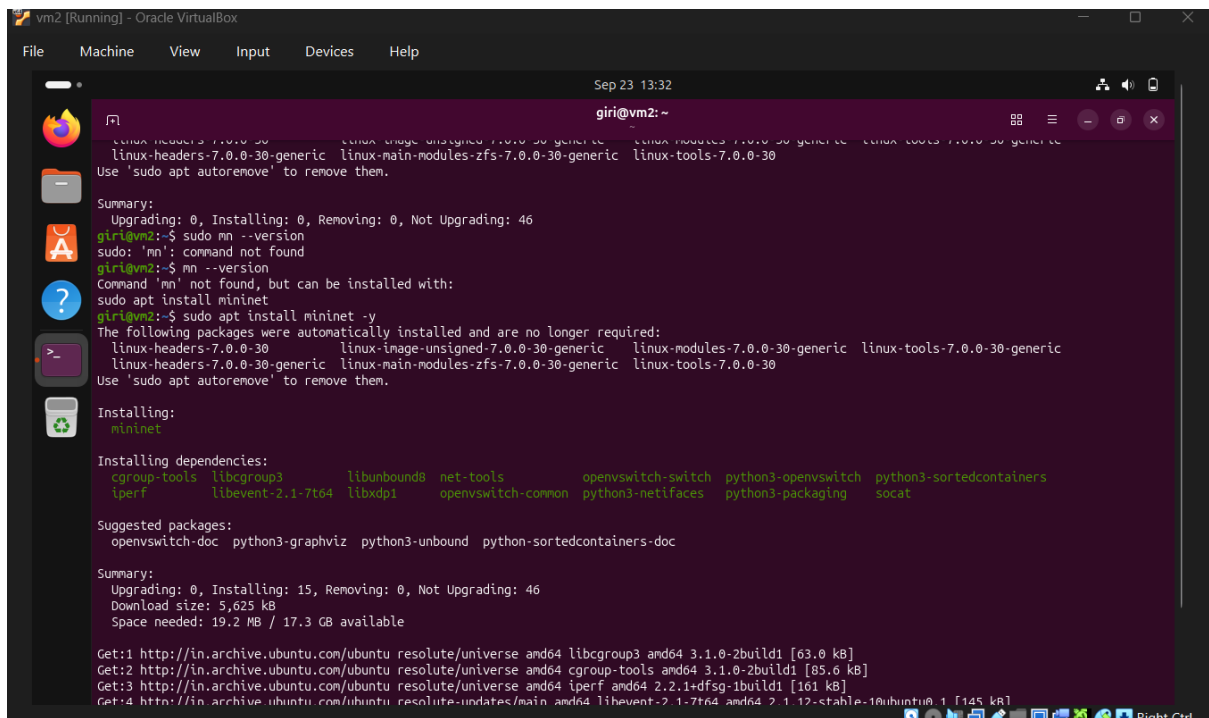
Objective

To create and configure a simple network topology using an open-source network virtualization tool and verify communication between the virtual network nodes.

Task

- Ubuntu Linux Virtual Machine
- VirtualBox
- Mininet
- Create simple Network Topology
- ping and basic Linux networking commands
- After completion, stop the network.

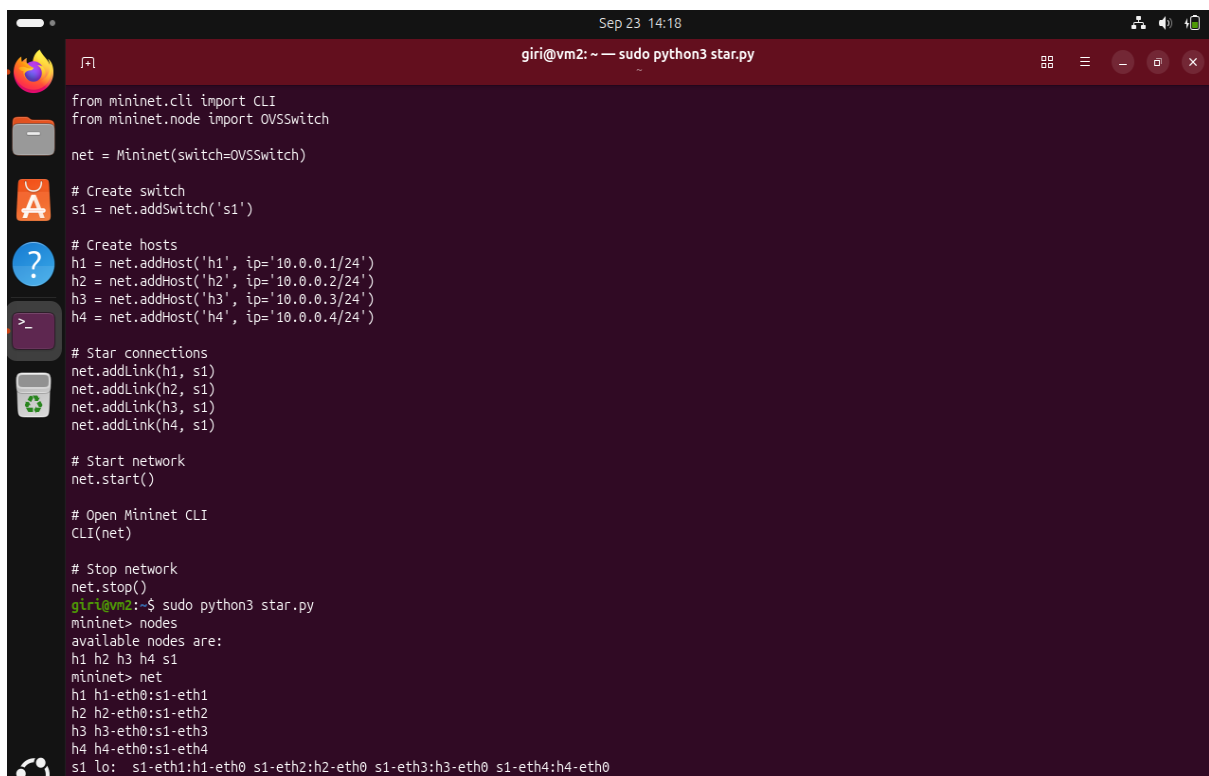
Mininet Installation



```
giri@vm2: ~  
$ sudo apt autoremove  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 46  
giri@vm2:~$ sudo mn --version  
sudo: 'mn': command not found  
giri@vm2:~$ mn --version  
Command 'mn' not found, but can be installed with:  
sudo apt install mininet  
giri@vm2:~$ sudo apt install mininet -y  
The following packages were automatically installed and are no longer required:  
linux-headers-7.0.0-30 linux-image-unsigned-7.0.0-30-generic linux-modules-7.0.0-30-generic linux-tools-7.0.0-30-generic  
linux-headers-7.0.0-30-generic linux-main-modules-zfs-7.0.0-30-generic linux-tools-7.0.0-30  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
mininet  
  
Installing dependencies:  
cgroup-tools libgroup3 libunbound8 net-tools openvswitch-switch python3-openvswitch python3-sortedcontainers  
iperf libevent-2.1-7t64 libxdp1 openvswitch-common python3-netifaces python3-packaging socat  
  
Suggested packages:  
openvswitch-doc python3-graphviz python3-unbound python-sortedcontainers-doc  
  
Summary:  
Upgrading: 0, Installing: 15, Removing: 0, Not Upgrading: 46  
Download size: 5,625 kB  
Space needed: 19.2 MB / 17.3 GB available  
  
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 libgroup3 amd64 3.1.0-2build1 [63.0 kB]  
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 cgroup-tools amd64 3.1.0-2build1 [85.6 kB]  
Get:3 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 iperf amd64 2.2.1+dfsg-1build1 [161 kB]  
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libevent-2.1-7t64 amd64 2.1.12-stable-10ubuntu0.1 [145 kB]
```

Topology

1.Star Topology



```
giri@vm2: ~ -- sudo python3 star.py  
from mininet.cli import CLI  
from mininet.node import OVSSwitch  
  
net = Mininet(switch=OVSSwitch)  
  
# Create switch  
s1 = net.addSwitch('s1')  
  
# Create hosts  
h1 = net.addHost('h1', ip='10.0.0.1/24')  
h2 = net.addHost('h2', ip='10.0.0.2/24')  
h3 = net.addHost('h3', ip='10.0.0.3/24')  
h4 = net.addHost('h4', ip='10.0.0.4/24')  
  
# Star connections  
net.addLink(h1, s1)  
net.addLink(h2, s1)  
net.addLink(h3, s1)  
net.addLink(h4, s1)  
  
# Start network  
net.start()  
  
# Open Mininet CLI  
CLI(net)  
  
# Stop network  
net.stop()  
giri@vm2:~$ sudo python3 star.py  
mininet> nodes  
available nodes are:  
h1 h2 h3 h4 s1  
mininet> net  
h1 h1-eth0:s1-eth1  
h2 h2-eth0:s1-eth2  
h3 h3-eth0:s1-eth3  
h4 h4-eth0:s1-eth4  
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0 s1-eth3:h3-eth0 s1-eth4:h4-eth0
```

2. BUS TOPOLOGY

```
giri@vm2:~$ sudo python3 bus.py
[sudo: authenticate] Password:

===== BUS TOPOLOGY =====
h1      h2      h3      h4
|        |        |        |
s1-----s2-----s3-----s4

mininet> nodes
available nodes are:
h1 h2 h3 h4 s1 s2 s3 s4
mininet> net
h1 h1-eth0:s1-eth2
h2 h2-eth0:s2-eth3
h3 h3-eth0:s3-eth3
h4 h4-eth0:s4-eth2
s1 lo: s1-eth1:s2-eth1 s1-eth2:h1-eth0
s2 lo: s2-eth1:s1-eth1 s2-eth2:s3-eth1 s2-eth3:h2-eth0
s3 lo: s3-eth1:s2-eth2 s3-eth2:s4-eth1 s3-eth3:h3-eth0
s4 lo: s4-eth1:s3-eth2 s4-eth2:h4-eth0
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet>
```

3. RING TOPOLOGY

```
giri@vm2:~$ sudo python3 ring.py

ip link show | egrep -o '([_.:alnum:]]+eth[[:digit:]]+)'
ip link show
*** Killing stale mininet node processes
kill -9 -f mininet:
*** Shutting down stale tunnels
kill -9 -f Tunnel-Ethernet
kill -9 -f .ssh/mn
rm -f ~/.ssh/mn/*
*** Cleanup complete.
giri@vm2:~$ sudo python3 ring.py

===== RING TOPOLOGY =====

      s1
     / \
    /   \
   s4     s2
    \   /
     \ /
      s3

h1 -> s1
h2 -> s2
h3 -> s3
h4 -> s4

mininet> nodes
available nodes are:
h1 h2 h3 h4 s1 s2 s3 s4
mininet> net
h1 h1-eth0:s1-eth3
h2 h2-eth0:s2-eth3
h3 h3-eth0:s3-eth3
h4 h4-eth0:s4-eth3
s1 lo: s1-eth1:s2-eth1 s1-eth2:s4-eth2 s1-eth3:h1-eth0
s2 lo: s2-eth1:s1-eth1 s2-eth2:s3-eth1 s2-eth3:h2-eth0
s3 lo: s3-eth1:s2-eth2 s3-eth2:s4-eth1 s3-eth3:h3-eth0
s4 lo: s4-eth1:s3-eth2 s4-eth2:s1-eth2 s4-eth3:h4-eth0
mininet> ping
```

4. TREE TOPOLOGY

```
Sep 23 17:22
giri@vm2: ~ — sudo python3 tree.py

giri@vm2:~$ nano tree.py
giri@vm2:~$ sudo python3 tree.py
[sudo: authenticate] Password:

===== TREE TOPOLOGY =====

      s1
     /  \
    s2   s3
   /  \ /  \
  h1  h2 h3  h4

mininet> nodes
available nodes are:
h1 h2 h3 h4 s1 s2 s3
mininet> net
h1 h1-eth0:s2-eth2
h2 h2-eth0:s2-eth3
h3 h3-eth0:s3-eth2
h4 h4-eth0:s3-eth3
s1 lo: s1-eth1:s2-eth1 s1-eth2:s3-eth1
s2 lo: s2-eth1:s1-eth1 s2-eth2:h1-eth0 s2-eth3:h2-eth0
s3 lo: s3-eth1:s1-eth2 s3-eth2:h3-eth0 s3-eth3:h4-eth0
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet>
```

5.MESH TOPOLOGY

```
Sep 23 17:31
giri@vm2: ~ — sudo python3 mesh.py

giri@vm2:~$ nano mesh.py
giri@vm2:~$ sudo python3 mesh.py

===== FULL MESH TOPOLOGY =====

      s1
     /  \
    s2---s3
     \  /
      s4

h1 -> s1
h2 -> s2
h3 -> s3
h4 -> s4

mininet> nodes
available nodes are:
h1 h2 h3 h4 s1 s2 s3 s4
mininet> net
h1 h1-eth0:s1-eth4
h2 h2-eth0:s2-eth4
h3 h3-eth0:s3-eth4
h4 h4-eth0:s4-eth4
s1 lo: s1-eth1:s2-eth1 s1-eth2:s3-eth1 s1-eth3:s4-eth1 s1-eth4:h1-eth0
s2 lo: s2-eth1:s1-eth1 s2-eth2:s3-eth2 s2-eth3:s4-eth2 s2-eth4:h2-eth0
s3 lo: s3-eth1:s1-eth2 s3-eth2:s2-eth2 s3-eth3:s4-eth3 s3-eth4:h3-eth0
s4 lo: s4-eth1:s1-eth3 s4-eth2:s2-eth3 s4-eth3:s3-eth3 s4-eth4:h4-eth0
mininet> pingall
*** Ping: testing ping reachability
h1 -> X X X
h2 -> X X X
h3 -> X X X
h4 -> X X X
*** Results: 100% dropped (0/12 received)
mininet>
```